

Alikasim Budhwani

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SKILLS

- **Programming Languages:** Python, C++, MATLAB, R, Bash
- **Toolboxes/Environments:** ROS, Linux, Docker, PyTorch, Git, MuJoCo

EXPERIENCE

- **BHF Robotics** [🌐] Sep 2023 - Dec 2024
Robotics Engineer Markham, Canada
 - Led a team of 4 in developing ROS-based autonomy stack for Agrobot mobile platform which conducts targeted weed elimination via high-voltage electricity.
 - Implemented ant colony optimization (ACO) for weed elimination sequencing, achieving 300 weeds/min cycle time.
 - Developed the perception module for precise weed detection and elimination site selection, minimizing crop damage while achieving 90% accuracy.
- **Ansys** [🌐] Jan 2023 - Apr 2023
SDE Intern Waterloo, Canada
 - Implemented isoparametric mapping in C++-based finite element solver, reducing computation time by 10% for 10,000+ node models with complex curved geometries.
 - Refactored C++ element classes in coupling framework using templates, eliminating 15+ redundant classes and improving maintainability.
 - Integrated unit tests (C++) in CI/CD pipeline to verify the functionality, and edge cases of higher-order finite mesh elements up to order 4.
- **Kinectrics** [🌐] May 2022 - Aug 2022
R&D Intern Etobicoke, Canada
 - Developed end-to-end robotic solutions for waste decommissioning of irradiated pressure vessels at Pickering Nuclear Station.
 - Researched, and validated UAV solutions for radiation monitoring and line inspection pilot projects.
- **Promation** [🌐] Sep 2021 - Dec 2021
Perception Intern Oakville, Canada
 - Developed an in-situ defect detection system for 3D metal printing, obtaining 75% detection accuracy.
 - Implemented a custom hand-eye calibration algorithm for 2D laser scanners, reaching 0.5mm accuracy during 3D object reconstruction.
 - Designed a vision-based trajectory adjustment algorithm for automotive robots, achieving 0.2mm recalibration error.
 - Created a semi-supervised data processing pipeline to segment printed samples, filter noise, and annotate point clouds, reducing time by 60%.

EDUCATION

- **University of Waterloo** Jan 2025 - Apr 2027
MASc in Mechanical Engineering (GPA: 4.00/4.00) Waterloo, Canada
 - **Research:** Leveraging chaotic dynamics to learn safety filters for contact-rich manipulation policies.
 - **Coursework** including Nonlinear Control, Stochastic Control, and Vehicle Dynamics.
 - Recipient of Ontario Graduate Scholarship (OGS).
- **University of Waterloo** Sep 2019 - Aug 2024
BASc in Mechanical Engineering (GPA: 3.84/4.00) Waterloo, Canada
 - **Coursework** including Robot Manipulation, Mobile Robotics, Reinforcement Learning and Multivariable Control.